Sparse Bayesian Multitask Learning for Radar Target Recognition

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Abstract: A multitask learning framework is proposed to jointly model the tasks by sharing the subspace in this paper. Firstly, we assume the tasks (the classification of one class versus others) are related. Then the task predictors may belong to a low dimensional subspace to realize the information sharing. The subspace bases of each task predictor can be learned by utilizing the Bernoulli-Beta prior from the given data, and the number of shared bases between different tasks can be utilized to automatically determine their relatedness. The proposed method is validated on the measured radar data.

Keywords: sparse; Bayesian; multitask learning; target recognition

I. INTRODUCTION

Radar high resolution range profile (HRRP) [1] is the amplitude of coherent summations of the complex time return from target scatters in each range resolution cell, which contains the target structure signatures, such as target size, scatterer distribution, etc.. Thus, HRRP is widely used in radar automatic target recognition.

In the practical applications, it is usually hard to collect enough training radar data, due to the sampling rate limitation or the noncooperative target identification. In such cases, multitask learning [2] between task domains would be desirable, which aims to improve generalization performance of related tasks by jointing learning.

In this paper, the proposed model is suitable for multicategory classification, in which each task is the classification of one class versus others. To realize the multitask learning, we assume the information is shared across tasks - the task predictors (classification weights) may belong to a subspace. However, if tasks are not related or related minimally, joint learning may degrade the classification performance. Thus, we utilize the number of shared bases in the subspace to represent the relatedness between the tasks, which provides the flexibility of no sharing when they are unrelated and partial/total sharing when they are related.

II. BASIC MODEL CONSTRUCTION

First we present the multi-task learning framework in this section. Suppose we have T tasks, indexed as $t=1,\cdots,T$. If $T=1$, it is the binary classification problem, if $T>2$, it is the multicategory classification problem, and each task is a binary classification problem. The training input-label pairs of the t th task denoted as $D_t=\{(\boldsymbol{x}_{ti},y_{ti})\,|\,i=1,\cdots,N_t\}$, where $\boldsymbol{x}_{ti}\in\mathbb{R}^{P\times 1}$ is the input sample, and $X_t=[\boldsymbol{x}_{t1},\cdots,\boldsymbol{x}_{tN_t}]$, $\boldsymbol{y}_t=[y_{t1},\cdots y_{tN_t}]^{\mathrm{T}}$ are the labels, and $y_{ti}\in\{+1,-1\}$.

In this paper, the latent variable representation of regularized support vector machine [3] (LVSVM) is utilized to realize the binary classification of each task, which is proposed by Polson based on data augmentation technique. By introducing the latent variables, LVSVM can be expressed in a Bayesian framework. Therefore, the Bayesian estimating algorithms can be utilized to obtain the estimates of SVM coefficients. Following, we will introduce the LVSVM briefly.

For the binary classification of each task, the discriminant function of the simple linear support vector machine (SVM) [4] is :

$$y_{ti}=\mathrm{sign}(\tilde{\boldsymbol{x}}_{ti}{}^{\mathrm{T}}\boldsymbol{\theta}_t)\ , \tag{1}$$

where $\tilde{\boldsymbol{x}}_{ti}=[1;\boldsymbol{x}_{ti}]$ is the augmented feature vector, and $\mathrm{sign}(\cdot)$ represents the sign function. Satisfies $\boldsymbol{x}_{ti}{}^{\mathrm{T}}\boldsymbol{\theta}_t>0$ for samples having $y_{ti}=+1$ and $\boldsymbol{x}_{ti}{}^{\mathrm{T}}\boldsymbol{\theta}_t<0$ for samples having $y_{ti}=-1$, so that $y_{ti}\boldsymbol{x}_{ti}{}^{\mathrm{T}}\boldsymbol{\theta}_t>0$ for all training data samples. To obtain the maximum margin and minimum misclassification rate, the task predictor $\boldsymbol{\theta}_t\in\mathbb{R}^{L\times 1}$ with $L=P+1$ is optimized to satisfy the objective function:

$$\min_{\boldsymbol{\theta}_t} d(\boldsymbol{\theta}_t)=\sum_{i=1}^{N_t}\max(1-y_{ti}\tilde{\boldsymbol{x}}_{ti}{}^{\mathrm{T}}\boldsymbol{\theta}_t,0)+\varphi(\boldsymbol{\theta}_t)\ , \tag{2}$$

where $\sum_{i=1}^{N_t}\max(1-y_{ti}\boldsymbol{x}_{ti}{}^{\mathrm{T}}\boldsymbol{\theta}_t,0)$ is the loss function, and $\varphi(\boldsymbol{\theta}_t)$ represents the constraint of the task predictor, which will be described in detail later on.

Reference [3] considers that minimizing equation (2) is equivalent to finding the mode of the pseudo-posterior distribution $p(\boldsymbol{\theta}_t\,|\,\boldsymbol{y}_t)$ defined by:

$$p(\boldsymbol{\theta}_t\,|\,\boldsymbol{y}_t)\propto\exp(-d(\boldsymbol{\theta}_t))\propto CL(\boldsymbol{y}_t\,|\,\boldsymbol{\theta}_t)p(\boldsymbol{\theta}_t)\ , \tag{3}$$

where C denotes the pseudo-posterior normalization constant, $L(\boldsymbol{y}_t \mid \boldsymbol{\theta}_t) = \prod_i L_i(\boldsymbol{y}_{ti} \mid \boldsymbol{\theta}_t) = \exp\{-2\sum_{i=1}^{N_t} \max(1 - y_{ti}\boldsymbol{x}_{ti}^{\mathrm{T}}\boldsymbol{\theta}_t, 0)\}$ represents the pseudo-likelihood, and $p(\boldsymbol{\theta}_t)$ is the prior distribution of the task predictor.

By introducing a latent variable $\beta_{ti} > 0$, reference [3] expresses the pseudo-likelihood contribution from observation $\boldsymbol{y}_{ti}$ as:

$$\begin{aligned} L_i(\boldsymbol{y}_{ti} \mid \boldsymbol{\theta}_t) &= \exp\{-2\max(1 - y_{ti}\boldsymbol{x}_{ti}^{\mathrm{T}}\boldsymbol{\theta}_t, 0)\} \\ &= \int_0^\infty \frac{1}{\sqrt{2\pi\beta_{ti}}} \exp(-\frac{1}{2}\frac{(1+\beta_{ti} - y_{ti}\boldsymbol{x}_{ti}^{\mathrm{T}}\boldsymbol{\theta}_t)^2}{\beta_{ti}}) d\beta_{ti} \end{aligned} \quad (4)$$

Since $L_i(\boldsymbol{y}_{ti} \mid \boldsymbol{\theta}_t)$ can also be expressed as $\int_0^\infty p(\boldsymbol{y}_{ti}, \beta_{ti} \mid \boldsymbol{\theta}_t) d\beta_{ti}$, then:

$$p(\boldsymbol{y}_{ti}, \beta_{ti} \mid \boldsymbol{\theta}_t) = \frac{1}{\sqrt{2\pi\beta_{ti}}} \exp(-\frac{1}{2}\frac{(1+\beta_{ti} - y_{ti}\boldsymbol{x}_{ti}^{\mathrm{T}}\boldsymbol{\theta}_t)^2}{\beta_{ti}}) p \quad (5)$$

Because of the latent variable β_{ti}, the binary classification of each task can be implemented by the Bayesian estimation methods, eg. Markov chain Monte Carlo (MCMC) method [5] and Variational Bayesian (VB) inference method [6] etc.

In traditional SVM, the popular constraint of task predictor is $\frac{1}{2}\|\boldsymbol{\theta}_t\|_2^2$. However, in order to realize the joint modeling (multitask learning) of the T tasks, here we adopt an alternative constraint - hierarchical factor analysis [7], which can model the data from T tasks through a subspace. Then some of the subspace bases are shared across tasks, while the others are individual to a task. Following is the hierarchical factorial leaning model of the t th task predictor:

$$\boldsymbol{\theta}_t = \boldsymbol{A}\boldsymbol{w}_t + \boldsymbol{\varepsilon}_t, \quad \boldsymbol{w}_t = \hat{\boldsymbol{w}}_t \odot \boldsymbol{z}_t \quad (6)$$

$$\begin{aligned} &\hat{\boldsymbol{w}}_t \sim \mathbb{N}(0, \alpha_t^{-1}\boldsymbol{I}_K), \\ &\quad \alpha_t \sim \mathrm{Gamma}(a_0, b_0) = \Gamma(a_0)^{-1} b_0^{a_0} \alpha_t^{a_0 - 1} \mathrm{e}^{-b_0\alpha_t} \end{aligned} \quad (7)$$

$$\boldsymbol{\varepsilon}_t \sim \mathbb{N}(0, \tau_t^{-1}\boldsymbol{I}_L), \quad \tau_t \sim \mathrm{Gamma}(c_0, d_0) \quad (8)$$

$$\boldsymbol{A} \sim \prod_{k=1}^K \mathbb{N}(0, \frac{1}{L}\boldsymbol{I}_L) \quad (9)$$

$$\boldsymbol{z}_t \sim \prod_{k=1}^K \mathrm{Bernoulli}(u_{tk}) = \prod_{k=1}^K u_{tk}^{z_{tk}} (1 - u_{tk})^{1 - z_{tk}} \quad (10)$$

$$\boldsymbol{u}_t \sim \prod_{k=1}^K \mathrm{Beta}(\frac{e_0}{K}, \frac{f_0(K-1)}{K}) \quad (11)$$

where $\boldsymbol{A} = [\boldsymbol{A}_1, \cdots, \boldsymbol{A}_k, \cdots, \boldsymbol{A}_K] \in \mathbb{R}^{L\times K}$ represents the shared subspace or dictionary of the T task predictors with $\boldsymbol{A}_k$ denoting the k th column of $\boldsymbol{A}$, K is an integer chosen to be the number of anticipated factors, $\boldsymbol{\varepsilon}_t \in \mathbb{R}^{L\times 1}$ represents the modeling error, $\boldsymbol{w}_t = [w_{t1}, \cdots, w_{tk}, \cdots, w_{tK}]^{\mathrm{T}} \in \mathbb{R}^{K\times 1}$ is the representation of the t th task predictor in the subspace, which has two components: subspace representation $\hat{\boldsymbol{w}}_t = [\hat{w}_{t1}, \cdots, \hat{w}_{tk}, \cdots, \hat{w}_{tK}]^{\mathrm{T}}$ and binary valued assignments $\boldsymbol{z}_t = [z_{t1}, \cdots, z_{tk}, \cdots, z_{tK}]^{\mathrm{T}}$, the symbol $\odot$ represents the Hadamard product or element-wise multiplication of two vectors. $\boldsymbol{z}_t \in \{0,1\}$ is a binary vector that encodes which subspace bases are activated for the corresponding task predictor and realizes the selection of the number of the factors automatically. We use $\mathbb{N}(\cdot)$, $\mathrm{Gamma}(\cdot)$, $\mathrm{Bernoulli}(\cdot)$ and $\mathrm{Beta}(\cdot)$ to refer respectively to Gaussian, Gamma, Bernoulli and Beta densities parameterized by the arguments inside parentheses. This is a hierarchical model, and the priors for the parameters of the density of $\boldsymbol{\theta}_t \mid \boldsymbol{A}, \hat{\boldsymbol{w}}_t, \boldsymbol{z}_t, \boldsymbol{\varepsilon}_t$ have their own hyper-priors with hyper-parameters. In (10) and (11), as K tends to infinity, the finite Bernoulli-Beta approximation approaches the Beta process [7]. If the truncation is large enough, data analyzed with this prior will exhibit fewer than K components.

With the prior of each parameter, the model likelihood can be expressed as

$$\begin{aligned} &\mathrm{p}(Y, \boldsymbol{\beta}, \boldsymbol{\Theta}, \boldsymbol{A}, \hat{\boldsymbol{w}}, \boldsymbol{\alpha}, \boldsymbol{z}, \boldsymbol{u}, \boldsymbol{\tau}) \\ &\quad = \prod_{t=1}^T \prod_{i=1}^{N_t} \mathrm{p}(\boldsymbol{y}_{ti}, \beta_{ti} \mid \boldsymbol{\theta}_t) \times \prod_{t=1}^T \mathbb{N}(\boldsymbol{\theta}_t \mid \boldsymbol{A}(\hat{\boldsymbol{w}}_t \odot \boldsymbol{z}_t), \tau_t^{-1} I_L) \\ &\quad \times \prod_{t=1}^T \mathbb{N}(\hat{\boldsymbol{w}}_t; 0, \alpha_t^{-1} I_K) \times \mathrm{Gamma}(\alpha_t; a_0, b_0) \times \prod_{k=1}^K \mathbb{N}(\boldsymbol{A}_k; 0, \frac{1}{L} I_L) \\ &\quad \times \prod_{t=1}^T \prod_{k=1}^K \mathrm{Bernoulli}(z_{tk}; u_{tk}) \times \mathrm{Beta}(u_{tk}; \frac{e_0}{K}, \frac{f_0(K-1)}{K}) \end{aligned} \quad (12)$$

For further illustration, we present the corresponding complete graphical model in Fig 1.

(8)

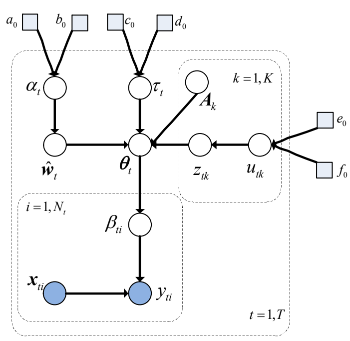

Fig. 1. Graphical model of the proposed multitask learning model

III. VARIATIONAL BAYESIAN INFERENCE

Bayesian inference seeks to estimate the posterior distribution of the latent variables $\boldsymbol{\Psi}$, given the observed data $\boldsymbol{D}$ and hyper-parameters Υ. Here we employ VB inference [6] as a compromise between accuracy and efficiency. In the proposed model, $\boldsymbol{D}$ is $\{\boldsymbol{x}_{ti}, y_{ti}\}_{t=1i=1}^{T\ N_t}$, $\boldsymbol{\Psi}$ is $\{\beta_{ti}, \boldsymbol{\theta}_t, \boldsymbol{A}, \hat{\boldsymbol{w}}_t, \alpha_t, \boldsymbol{z}_t, u_{tk}, \tau_t\}_{t=1i=1k=1}^{T\ N_t\ K}$, Υ is $\{a_0, b_0, c_0, d_0, e_0, f_0\}$. The VB inference for the model is summarized in Algorithm 1, in which $\langle\cdot\rangle$ denotes the expectation, $\mathcal{IG}(\cdot)$ represents the inverse Gaussian distribution [3], and $\mathcal{IG}(\beta;\lambda,\delta)=\sqrt{\frac{\delta}{2\pi\beta^3}}\exp(-\frac{\delta(\beta-\lambda)^2}{2\lambda^2\beta})$, then the mean and variance of variable β are $\langle\beta\rangle=\lambda$ and $Var(\beta)=\lambda^3/\delta$.

Algorithm 1: VB-based Inference for the Proposed Model

Input: the input-label pairs of each task $\boldsymbol{D}=\{\boldsymbol{x}_{ti}, y_{ti}\}_{t=1i=1}^{T\ N_t}$; the initial hyper-parameters Υ_0: $a_0=b_0=c_0=d_0=10^{-6}$, $e_0=f_0=1$; $K=50$; a small positive constant $\varsigma=10^{-6}$。

Output: the posterior distributions of the latent variables $\boldsymbol{\Psi}$: $\mathrm{q}(\beta_{ti})$, $\mathrm{q}(\boldsymbol{\theta}_t)$, $\mathrm{q}(A_{kl})$, $\mathrm{q}(\hat{\boldsymbol{w}}_t)$, $\mathrm{q}(z_{tk})$, $\mathrm{q}(u_{tk})$, $\mathrm{q}(\alpha_t)$, $\mathrm{q}(\tau_t)$。

Initialization: $\Upsilon \leftarrow \Upsilon_0$; $iter=0$; $change=1$; $\mathrm{L}(1)=\int \mathrm{q}(\boldsymbol{\Psi})\ln\frac{\mathrm{p}(\boldsymbol{D},\boldsymbol{\Psi})}{\mathrm{q}(\boldsymbol{\Psi})}\mathrm{d}\boldsymbol{\Psi}$。

While $change >= \varsigma$

$iter = iter+1$

$\mathrm{q}(\beta_{ti}^{-1})=\mathcal{IG}(\lambda_{ti},\delta_{ti});\ \lambda_{ti}=\left|1-y_{ti}\boldsymbol{x}_{ti}^{\mathrm{T}}\langle\boldsymbol{\theta}_t\rangle\right|^{-1},\ \delta_{ti}=1$

$\mathrm{q}(\boldsymbol{\theta}_t)=\mathbb{N}(\xi_t,\Lambda_t)$; $\Lambda_t=(\sum_{i=1}^{N_t}\frac{y_{ti}^{\ 2}}{\langle\beta_{ti}\rangle}\boldsymbol{x}_{ti}\boldsymbol{x}_{ti}^{\mathrm{T}}+\langle\tau_t\rangle\mathrm{I}_L)^{-1}$,

$\xi_t=\Lambda_t(\sum_{i=1}^{N_t}(1+\langle\beta_{ti}\rangle^{-1})y_{ti}\boldsymbol{x}_{ti}+\langle\tau_t\rangle\langle\boldsymbol{A}\rangle diag\langle\boldsymbol{z}_t\rangle\langle\hat{\boldsymbol{w}}_t\rangle)$

$\mathrm{q}(\hat{\boldsymbol{w}}_t)=\mathbb{N}(\boldsymbol{\eta}_t,\boldsymbol{\Sigma}_t)$; $\boldsymbol{\Sigma}_t=(\langle\alpha_t\rangle\boldsymbol{I}_K+\langle\tau_t\rangle\langle\boldsymbol{z}_t\boldsymbol{z}_t^{\mathrm{T}}\rangle\odot\langle\boldsymbol{A}^{\mathrm{T}}\boldsymbol{A}\rangle)^{-1}$,

$\boldsymbol{\eta}_t=\boldsymbol{\Sigma}_t\langle\tau_t\rangle diag(\langle\boldsymbol{z}_t\rangle)\langle\boldsymbol{A}^{\mathrm{T}}\rangle\langle\boldsymbol{\theta}_t\rangle)$

$\mathrm{q}(\alpha_t)=\mathrm{Gamma}(\tilde{a}_t,\tilde{b}_t);\ \tilde{a}_t=a_0+K/2,\ \tilde{b}_t=b_0+\frac{1}{2}\langle\hat{\boldsymbol{w}}_t^{\mathrm{T}}\hat{\boldsymbol{w}}_t\rangle$

$\mathrm{q}(A_{kl})=\mathrm{CN}(\zeta_{kl},B_{kl});\ B_{kl}=(L+\sum_{t=1}^{T}\langle\tau_t\rangle\langle z_{tk}^{\ 2}\rangle\langle\hat{w}_{tk}^{\ 2}\rangle)^{-1}$,

$\zeta_k=\boldsymbol{B}_k(\sum_{t=1}^{T}\langle\tau_t\rangle\langle z_{tk}\rangle\langle\hat{w}_{tk}\rangle(\langle\boldsymbol{\theta}_t\rangle-\sum_{h\neq k}^{K}\langle z_{th}\rangle\langle\hat{w}_{th}\rangle\boldsymbol{A}_h))$

$\mathrm{q}(\tau_t)=\mathrm{Gamma}(\tilde{c}_t,\tilde{d}_t);\quad \tilde{c}_t=c_0+L/2$,

$\tilde{d}_t=d_0+\frac{1}{2}(\langle\boldsymbol{\theta}_t^{\mathrm{T}}\boldsymbol{\theta}_t\rangle-2\langle\hat{\boldsymbol{w}}_t\rangle diag\langle\boldsymbol{z}_t\rangle\langle\boldsymbol{A}\rangle\langle\boldsymbol{\theta}_t\rangle+$
$\langle\hat{\boldsymbol{w}}_t^{\mathrm{T}}diag(\boldsymbol{z}_t)\boldsymbol{A}^{\mathrm{T}}\boldsymbol{A}diag(\boldsymbol{z}_t)\hat{\boldsymbol{w}}_t^{\mathrm{T}}\rangle)$

$\mathrm{q}(z_t)=\mathrm{Bernoulli}(\frac{\mathrm{p}(z_{tk}=1)}{\mathrm{p}(z_{tk}=0)+\mathrm{p}(z_{tk}=1)})$;

$\mathrm{p}(z_{tk}=0)\propto\exp(\langle\ln(1-u_{tk})\rangle)$,

$\mathrm{p}(z_{tk}=1)\propto\exp(\langle\ln(u_{tk})\rangle)\times$

$\exp(\langle\tau_t\rangle(\langle\hat{w}_{tk}\rangle\langle\boldsymbol{A}_k^{\mathrm{T}}\rangle(\langle\boldsymbol{\theta}_t\rangle-\sum_{h\neq k}^{K}z_{th}\langle\hat{w}_{th}\rangle\langle\boldsymbol{A}_h\rangle)-\frac{1}{2}\langle\hat{w}_{th}^{\ 2}\rangle\langle\boldsymbol{A}_k^{\mathrm{T}}\boldsymbol{A}_k\rangle))$

$\mathrm{q}(u_{tk})=\mathrm{Beta}(\frac{\tilde{e}_{tk}}{K},\frac{\tilde{f}_{tk}(K-1)}{K})$;

$\tilde{e}_{tk}=e_0+K\langle z_{tk}\rangle$, $\tilde{f}_{tk}=f_0+\frac{K}{K-1}(1-\langle z_{tk}\rangle)$

$\mathrm{L}(iter+1)=\int\mathrm{q}(\boldsymbol{\Psi})\ln\frac{\mathrm{p}(\boldsymbol{D},\boldsymbol{\Psi})}{\mathrm{q}(\boldsymbol{\Psi})}\mathrm{d}\boldsymbol{\Psi}$;

$change=(\mathrm{L}(iter{+}1)\text{-}\mathrm{L}(iter))/\mathrm{L}(iter)$

End

At convergence of the parameter estimation, we can obtain the posterior expectations $\langle\boldsymbol{\theta}_t\rangle$, $\langle\boldsymbol{A}\rangle$, $\langle\hat{\boldsymbol{w}}_t\rangle$, $\langle\boldsymbol{z}_t\rangle$ and $\langle\tau_t\rangle$. In addition, $\boldsymbol{z}^{(v,m)}$ is binary vector, if $\mathrm{p}(z_{tk}=1)\geq 0.5$, then $z_{tk}=1$, otherwise $z_{tk}=0$.

IV. PREDICTION

After accomplishing the model learning of the training stage, following, we present the classification of test sample $\boldsymbol{x}^*$. Since each task is a classifier, then we can put the test sample in each classifier in turn. By utilizing the leaned predictor $\langle\boldsymbol{\theta}_t\rangle$ of each task , the classification value va_t of each task can be obtained:

$$va_t=\langle\boldsymbol{\theta}_t\rangle^{\mathrm{T}}\boldsymbol{x}^*\ . \tag{13}$$

If $T=1$, then $y^*=sign(va_t)$, if $y^*=1$, $\boldsymbol{x}^*$ is determined as the first class, otherwise, it is classified as the other class. If $T>2$, the class label of $\boldsymbol{x}^*$ is $t^*=\arg\max_t va_t$.

V. EXPERIMENTAL RESULTS

In this section we present experimental results on the measured radar data set of three types of planes [1]. The parameters of radar and planes are given in Table 1, and the measured HRRP is a 256-dimensional vector. It is a prerequisite for HRRP with the target-aspect, time-shift, and amplitude-scale sensitivity problems [8]. Here, we use the aspect-frame partition method to deal with the target-aspect sensitivity [8]. To deal with the time-shift sensitivity, we extract the time-shift invariant feature - power spectrum [9], which has the symmetrical structure, then $P=129$. To deal with the amplitude-scale sensitivity, the L_2 amplitude normalization can be used [8]. Since the measured HRRP data only contains three categories of planes, thus, to validate the ability of multicategory classification of the proposed method, three aspect-frames with great difference of each category are regarded as three categories, which is based on the target-aspect sensitivity of the HRRP data. Then, we have nine categories in all, i.e. $T = 9$ .And each category has 1024 samples, in which 524 fixed samples are test samples, the remaining 500 samples are the training samples. Thus, we have 4721 test samples in all.

TABLE I. PARAMETER OF PLANES AND RADAR IN THE ISAR EXPERIMENT

radar parameters	center frequency		5520 MHz
	bandwidth		400 MHz
planes	length (m)	width	height
Yak-42	36.38	34.88	9.83
Cessna Citation	14.40	15.90	4.57
An-26	23.80	29.20	9.83

Following, we will investigate how the number of training samples influences the recognition performance of the proposed method, Support Vector Machine (SVM) (including linear SVM and kernel SVM) [4] and Adaptive Gaussian Classifier (AGC) [10], which are applicable to the small sample target recognition . The training samples are randomly selected from the 500-example training set of each category. The number of the training samples of each category increases from 10 to 100 (the interval is 10). Fig. 2 shows the recognition performance of the competing algorithms versus the number of training. From Fig.2, we can see that the average recognition rate of AGC is lowest among the four classifiers. Although linear SVM (LSVM) performs better than AGC, its recognition rate is also lower than kernel SVM (KSVM) and our proposed method. When the number of the training data per category is smaller than 50, our method performs better than kernel SVM, when the training data number is greater than 50, our method is comparable to kernel SVM, which demonstrates that our method is more robust to the change of the training data number compared with kernel SVM.

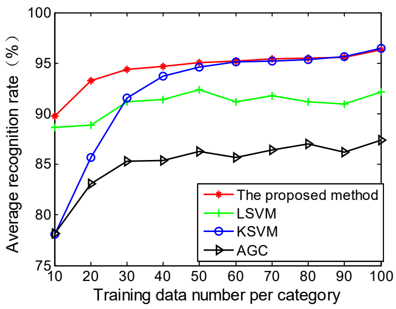

Fig. 2. Variation of the recognition performance with training data size

Since the information sharing among the task predictors is realized by sharing the same subspace, and the relatedness between tasks is determined by the number of the shared bases. Therefore, the binary vector $\langle z_t \rangle$ of each task predictor is given in Fig. 3. From Fig. 3, we can see that the number of nonzero elements of $\langle z_t \rangle$ is obviously smaller than the anticipated $K = 50$, which justifies the automatic inference of the number of the bases of each task by using the Bernoulli-Beta prior. In addition, the different tasks may share the same bases, or have their individual bases, which indicates the proposed method can learn the relatedness between tasks automatically based on the given data.

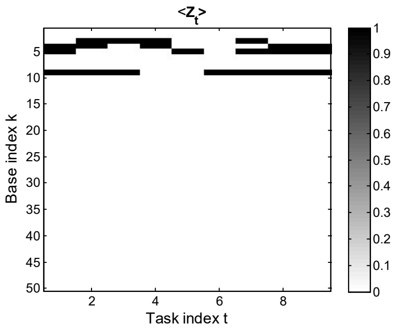

Fig. 3. Base usage of each task

VI. CONCLUSION

To guarantee the recognition performance when the training data size is small, we present a Bayesian multitask learning framework to jointly model the tasks with varying degree of relatedness. The relatedness between tasks can be automatically learned by utilizing the Bernoulli-Beta prior. The experimental results on the measured radar data demonstrate that our method can obtain good recognition performance.

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